InfraMitra — AI GPU Infrastructure & Data Center Operations Curriculum

Compressed 8-Week Industry-Ready Program

InfraMitra is a compressed hands-on infrastructure operations program designed to create deployment-ready technicians and junior operators for enterprise data centers, AI GPU clusters, HPC facilities, and cloud infrastructure environments. The program combines: • Enterprise infrastructure operations • AI GPU infrastructure exposure • HPC operational awareness • Real-world troubleshooting • Monitoring and operational workflows • Documentation and soft skills readiness

Phase	Duration	Focus
Foundation Infrastructure Operations	2 Weeks	Rack, power, cabling, deployment
Enterprise Infrastructure Operations	2 Weeks	Monitoring, operations, troubleshooting
AI GPU Infrastructure & HPC Operations	2 Weeks	GPU systems, AI clusters, AI workloads
Integrated Operations & Deployment Readiness	2 Weeks	Incidents, documentation, deployment readiness

Week 1 — Infrastructure Foundations

- Data center basics
- Rack and server identification
- GPU server awareness
- Receiving and unpacking procedures
- Fiber and power infrastructure basics
- Inventory and asset tagging

Week 2 — Rack Assembly & Cabling

- Rack assembly
- Structured cabling
- Fiber patching
- UPS and PDU basics
- BIOS and remote management
- Deployment workflows

Week 3 — Enterprise Operations & Monitoring

- NOC and data center operations
- Monitoring dashboards
- Environmental monitoring

- Incident management basics
- Ticketing systems
- Server health checks

Week 4 — Maintenance & Troubleshooting

- Preventive maintenance
- Power troubleshooting
- Cable diagnostics
- Hardware replacement
- Firmware awareness
- Incident escalation workflows

Week 5 — AI GPU Infrastructure Foundations

- CPU vs GPU architecture
- CUDA ecosystem overview
- GPU memory concepts
- NVLink awareness
- AI data center architecture
- HPC vs traditional data centers
- GPU rack density and cooling
- AI power consumption awareness

Week 6 — AI GPU Cluster Operations

- GPU clustering concepts
- AI workload scheduling
- GPU utilization monitoring
- AI load management
- Distributed training awareness
- GPU telemetry
- Thermal throttling awareness
- Kubernetes and container awareness
- AI cluster reliability concepts

Week 7 — Documentation & Incident Management

- Rack diagrams
- Asset documentation
- GPU inventory management

- Incident simulations
- Capacity planning awareness
- Operational reporting

Week 8 — Deployment Readiness

- Production simulation
- Monitoring exercises
- AI GPU operations
- Safe shutdown procedures
- Decommissioning workflows
- Final practical evaluation

Soft Skills & Professional Readiness

- Professional communication
- Shift handover discipline
- Incident reporting communication
- Team collaboration
- Problem-solving mindset
- Root cause analysis basics
- Documentation discipline
- Resume and interview preparation
- Customer and vendor communication awareness
- Operational etiquette and workplace discipline

Industry Reference Areas
• NVIDIA GPU infrastructure concepts
• CUDA ecosystem awareness
• NVIDIA DGX architecture awareness
• Microsoft Azure AI infrastructure concepts
• HPC operational fundamentals
• AI cluster operations awareness
• GPU monitoring and telemetry concepts

Program Outcomes
Students completing the program gain exposure to:

• Enterprise infrastructure operations
• GPU infrastructure awareness
• AI cluster operational workflows
• Monitoring and troubleshooting fundamentals
• Documentation and incident handling
• HPC vs traditional data center understanding
• Deployment-ready operational discipline

Source reference aligned with original InfraMitra curriculum and expanded using industry AI infrastructure operational concepts inspired by NVIDIA, Microsoft Azure AI infrastructure, HPC operations, and enterprise data center workflows.